CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 22: Heights and Distances

Every formula and result of the chapter on one sheet

1	Angle of elevation	Angle between the horizontal and the line of sight, object above the eye
2	Angle of depression	Angle between the horizontal and the line of sight, object below the eye
3	Elevation and depression	Angle of depression of B from A = angle of elevation of A from B
4	Height and distance	$\tan\theta = \dfrac{\text{height}}{\text{horizontal distance}}$
5	Height and slant length	$\sin\theta = \dfrac{\text{height}}{\text{slant length}}$
6	Distance and slant length	$\cos\theta = \dfrac{\text{horizontal distance}}{\text{slant length}}$
7	tan of standard angles	$\tan 30^\circ = \dfrac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$
8	sin and cos of standard angles	$\sin 30^\circ = \cos 60^\circ = \dfrac{1}{2}$; $\sin 60^\circ = \cos 30^\circ = \dfrac{\sqrt{3}}{2}$; $\sin 45^\circ = \cos 45^\circ = \dfrac{1}{\sqrt{2}}$
9	Objects on opposite sides	Distance between them $= d_1 + d_2$
10	Objects on the same side	Distance between them $= d_1 - d_2$
11	Useful values	$\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$